NOT MEASUREMENT SENSITIVE

MIL-PRF-81352D
23 December 2024
SUPERSEDING
MIL-PRF-81352C
16 June 2005

PERFORMANCE SPECIFICATION

COATINGS, AIRCRAFT TOUCH-UP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers general touch-up coatings for use on aircraft and support equipment. The volatile organic compound (VOC) content of these coatings is less than 340 grams/liter (g/l) (2.837 pounds/gallon).

1.2 Classification.

1.2.1 Types. The coatings are of the following types, as specified (see 6.2):

Type I	-	Acrylic base
Type II	-	Alkyd base
Type III	-	Polyurethane base
Type IV	-	Polysiloxane base

1.3 Part or identifying number (PIN). PINs to be used for coatings acquired to this specification are created as follows:

M81352	X	XX	XXXXX
Specification identification	Type designator Type I = 1 Type II = 2 Type III = 3 Type IV = 4	Container size designator 1 pint = 1P 1 quart = 1Q 1 gallon = 1G 1 Aerosol container = 1A 1 Touch-up kit = 1T	SAE AMS-STD-595 Color designation

Comments, suggestions, or questions on this document should be addressed to: Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to NAVAIR-Standardization@us.navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATION

TT-P-2760	-	Primer Coating: Polyurethane, Elastomeric, High-Solids

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-DTL-5624	-	Turbine Fuel, Aviation, Grade JP-5 (NATO F-44)
MIL-PRF-8625	-	Anodic Coatings for Aluminum and Aluminum Alloy
MIL-PRF-22750	-	Coating, Epoxy, High-Solids
MIL-PRF-23377	-	Primer Coatings: Epoxy, High-Solids
MIL-PRF-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number O-152, O-154, O-156, O-167.
MIL-PRF-32239	-	Coating System, Advanced Performance, for Aerospace Applications
MIL-DTL-53039	-	Camouflage Coating, Solvent Based, Chemical Agent Resistant
MIL-DTL-64159	-	Camouflage Coating, Water Dispersible Chemical Agent Resistant
MIL-DTL-81772	-	Thinner, Aircraft Coating
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-85285	-	Topcoat, Aircraft and Support Equipment
MIL-PRF-85582	-	Primer Coatings: Epoxy, Waterborne

(Copies of these documents are available at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection By Attributes

(Copies of this document are available from https://www.asq.org.)

ASTM INTERNATIONAL

ASTM D522/ D522M - Standard Test Method for Mandrel Bend Test of Attached Organic Coatings
ASTM D523 - Standard Test Method for Specular Gloss
ASTM D823 - Standard Practices for Producing Films of Uniform Thickness of Paint, Coatings, and Related Products on Test Panels
ASTM D1200 - Standard Test Method for Viscosity by Ford Viscosity Cup
ASTM D1210 - Standard Test Method for Fineness of Dispersion of Pigment-Vehicle Systems by Hegman-Type Gage
ASTM D1308 - Standard Test Method for Effect of Household Chemicals on Clear and Pigmented Coating Systems
ASTM D1640/D1640M - Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings
ASTM D2244 - Standard Practice for Calculation of Color Tolerances and Color Differences from Instrumentally Measured Color Coordinates
ASTM D2805 - Standard Test Method for Hiding Power of Paints by Reflectometry
ASTM D3335 - Standard Test Method for Low Concentrations of Lead, Cadmium and Cobalt in Paint by Atomic Absorption Spectroscopy
ASTM D3359 - Standard Test Methods for Rating Adhesion by Tape Test
ASTM D3718 - Standard Test Method for Low Concentrations of Chromium in Paint by Atomic Absorption Spectroscopy
ASTM D3924 - Standard Specification for Standard Environment for Conditioning and Testing of Paint, Varnish, Lacquer, and Related Materials
ASTM D3960 - Standard Practice for Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings
ASTM D5402 - Standard Practice for Assessing the Solvent Resistance of Organic Coatings Using Solvent Rubs
ASTM D5895 - Standard Test Methods for Evaluating Drying or Curing During Film Formation of Organic Coatings Using Mechanical Recorders
ASTM D6905 - Standard Practice for Impact Flexibility of Organic Coatings
ASTM G155 - Standard Practice for Operating Xenon Arc Lamp Apparatus for Exposure of Materials

(Copies of these documents are available from https://www.astm.org.)

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SAE INTERNATIONAL

SAE AMS3721	-	Tape for Organic Coatings Adhesion Testing
SAE AMS4035	-	Aluminum Alloy, Sheet and Plate, 4.4Cu-1.5Mg-0.60Mn (2024-O) Annealed; or when specified, "As Fabricated" (2024-F) UNS A92024
SAE AMS4037	-	Aluminum Alloy, Sheet and Plate 4.4Cu-1.5Mg-0.60Mn (2024; -T3 Flat Sheet, -T351 Plate) Solution Heat Treated - UNS A92024
SAE AMS-STD-595	-	Colors Used in Government Procurement

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Coatings furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.3 and 6.3).

3.2 Toxicity. The coatings shall have no adverse effect on the health of personnel when used for their intended purpose and with the precautions listed in 3.9.1 through 3.9.3, when evaluated in accordance with 4.6.12 (see 6.4). Components of the topcoats shall not contain cadmium, cadmium compounds, nor more than 0.05 percent by weight of lead metal or lead compounds. The topcoats shall not contain hexavalent chromium.

3.3 Composition. The topcoats furnished under this specification shall be a single component kit. The addition of a container of solvent is permissible as these coatings can have both brush and spray applications. Materials used to manufacture these topcoats are not limited, provided that the resulting topcoats meet the requirements of this specification (see 4.6).

3.3.1 Volatile organic compound (VOC) content. When mixed in accordance with the manufacturer's instructions to achieve the application viscosity specified in 3.5.3, the VOC content of the coatings shall be not greater than 340 g/l (2.837 lbs/gal), when tested in accordance with ASTM D3960 (see 4.6). Halogenated solvents, except for parachlorobenzyltrifluoride (PCBTF) or equivalent, as determined by the qualifying activity, shall be prohibited in the formulation of this product.

3.3.2 Thinner compatibility. The coatings shall be miscible in water or compatible with the following solvents: type I and II coatings shall be compatible with MIL-DTL-81772, type III or the manufacturers recommendations. Type III and IV coatings shall be compatible with MIL-DTL-81772, type I and IV or the manufacturers recommendations.

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3.4 Storage stability.

3.4.1 Storage stability. The unopened coating components, as packaged by the manufacturer, shall meet all requirements of this specification after storage for a period of one year for kits in plastic and metal containers greater than 120 ml (4 ounces) (see 4.6 and 6.4.3). Kits in plastic containers less than 120 ml (4 ounces) shall meet all requirements of this specification for six months (see 4.6 and 6.4.3). The conditions for the storage shall be a daily ambient air temperature that is maintained at 2 to 46 °C (35 to 115 °F).

3.4.2 Accelerated storage stability. The coatings, as packaged by the manufacturer, shall not develop increased internal container pressure when tested in accordance with 4.6.1. After storage, the contained material shall be free of lumps, skins, gels, or particulate matter, either suspended in solution or settled on the inner surface of the container. When stirred with a paddle, the contained material shall be a smooth, homogenous mixture. The coating when applied to a thickness of 1.7 to 2.3 mils [43-58 microns (µm)], shall dry tack-free within 60 minutes when tested in accordance to ASTM D5895 and shall have an adhesion rating of no less than 4A when examined in accordance with ASTM D3359.

3.5 Physical properties.

3.5.1 Condition in container. The coating shall exhibit no grit, skinning, curdling, livering or excessive pigment flotation. The coating shall show no more pigment settling or caking than can be easily and completely reincorporated to a smooth homogenous state by mixing with a hand paddle. In addition, the container shall exhibit no deformity (see 4.6.13).

3.5.2 Fineness of grind. The fineness of grind of the coatings shall be not less than 7.0 for gloss colors and not less than 5.0 for all other colors, when tested in accordance with ASTM D1210 (see 4.6).

3.5.3 Viscosity. The viscosity of the coatings, as prepared for application in accordance with the manufacturer's instructions, shall be 16 to 25 seconds through a #4 Ford cup, when tested in accordance with ASTM D1200 (see 4.6).

3.6 Working properties. The coatings shall yield a smooth and uniform film when applied by spray, brush, or roller and shall conform to 3.6.1 through 3.6.3.

3.6.1 Drying time. The coatings, when applied to a wet film thickness of 1.7 to 2.3 mils [43 to 58 microns (µm)], shall dry tack-free within 60 minutes, when tested in accordance with ASTM D5895 (see 4.6).

3.6.2 Surface appearance. The coatings shall dry to a smooth, uniform film, free of runs, sags, bubbling, wrinkling, streaking, blushing, haze, blisters, coarse particles, or other surface irregularity when prepared and examined in accordance with 4.6.2 (see 4.6).

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3.6.3 Coating system compatibility. All coatings supplied under this specification shall be compatible with the following coatings: MIL-PRF-22750, MIL-DTL-53039, MIL-DTL-64159, MIL-PRF-85285, MIL-PRF-23377, MIL-PRF-85582, TT-P-2760 and MIL-PRF-32239.

3.7 Film properties.

3.7.1 Color. When tested in accordance with ASTM D2244 (see 4.6), the color of the coatings shall be a visual match with the specified SAE AMS-STD-595 color chip (see 6.2).

3.7.2 Specular gloss. The applied coatings shall conform to table I when tested at a 60° geometry in accordance with ASTM D523 (see 4.6).

TABLE I. Specular gloss.

	Minimum	Maximum
Gloss	80	---
Semi-gloss	15	45
Camouflage	---	5

3.7.3 Adhesion. The topcoat shall have a rating of no less than 4A, when examined in accordance with ASTM D3359 (see 4.6.4).

3.7.4 Opacity (hiding power). The topcoat shall have a contrast ratio of no less than 0.95 for all colors except yellow (SAE AMS-STD-595, Color 13538), which shall have a contrast ratio of not less than 0.90, when cast to a dry film thickness of 1.9 to 2.1 mils [48 to 53 micrometer (μm)] on a black and white chart and tested in accordance with 4.6.5.

3.7.5 Flexibility.

3.7.5.1 Impact flexibility (type III and IV only). The type III and IV topcoats shall meet the following minimum requirement for ambient temperature impact flexibility: 20 percent, when tested in accordance with 4.6.6.1.

3.7.5.2 Mandrel bend flexibility. The coatings, applied to test panels in accordance with 4.5 through 4.5.2, shall exhibit no cracking, peeling, or loss of adhesion, when bent over a 1.0 in. (25.4 mm) mandrel, in accordance with ASTM D522/D522M, Method B (see 4.6.6.2).

3.8 Resistance properties.

3.8.1 Fluid resistance. The coatings, applied to test panels in accordance with 4.5 through 4.5.2, and tested in accordance with 4.6.7, shall exhibit no blistering, softening, or other coating defects four hours after removal from the fluids. A slight staining of the coatings is acceptable.

3.8.2 Weather resistance. Coating types I, II and III shall have a color difference value (ΔE) of not greater than 2.0, as compared to the unexposed coatings. The type IV coating shall not

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have a color difference value (ΔE) of not greater than 1.0, as compared to the unexposed coatings. All types of coating shall be tested in accordance with 4.6.8.

3.8.3 Heat resistance. The color change of the coatings (ΔE) shall be not greater than 1.0 as compared to the unexposed control coating, when tested in accordance with 4.6.9.

3.8.4 Solvent resistance (cure), (type III and IV). The topcoat shall withstand 25 double rubs with a cotton terrycloth rag soaked in methyl ethyl ketone (MEK) solvent (see 6.7), when tested in accordance with 4.6.10. Rubbing through to the primer indicates failure of the topcoat due to improper cure.

3.8.5 Water resistance. The coatings shall exhibit no cracking, blistering, or whitening when tested in accordance with 4.6.11. A slight whitening or dulling, which is removed by lightly wiping with a soft cloth, shall not be cause for rejection.

3.9 Identification of material. Individual containers greater than 1 pint and cases of containers 1 pint or less, shall bear permanent labels showing the information contained herein.

Specification MIL-PRF-81352 “Coatings, Aircraft Touch-up”
Type number and name (see 1.2.1)
Color (SAE AMS-STD-595 color name and number)
Manufacturer’s name, product identification, and lot number
Date of manufacture (month/year)
VOC content (g/l) and type of thinner compatible with the coating
Net contents
Mixing and thinning instructions

3.9.1 Component containers. All containers of toxic and hazardous chemicals and materials shall be labeled in accordance with the applicable federal, state, and municipal laws, statutes, regulations, and ordinances. In addition to the labeling, the following shall appear on every kit, on each exterior shipping container, and on the case of containers less than 1 pint.

CAUTION
THIS COATING MATERIAL IS TOXIC AND FLAMMABLE (if applicable)
DO NOT USE IN CONFINED AREAS
DO NOT USE WHERE THERE ARE OPEN FLAMES, ARCING EQUIPMENT, HOT SURFACES, NOR WHERE SMOKING IS PERMITTED.
USE ONLY WITH ADEQUATE VENTILATION.
AVOID BREATHING OF VAPOR.
DO NOT GET IN EYES, ON SKIN, ON CLOTHING.
IN CASE OF CONTACT, IMMEDIATELY FLUSH EYES OR SKIN WITH PLENTY OF WATER. FOR EYES, GET MEDICAL ATTENTION.

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3.9.2 Lid. Each can of material shall have a red printed label on its lid with the following information:

USE CAUTION.
OPEN SLOWLY

3.9.3 Precautions. Each kit shall include a sheet listing the following precautions:

a. The surface to be coated shall be clean (free of oil, dust, etc.)
b. Spray equipment shall be adequately grounded. Clean equipment with a compatible thinner immediately after use.
c. Apply to pretreated metal or appropriately prepared organic matrix composite surfaces.
d. Dispose of, without opening, any container that is bulged, deformed or otherwise compromised.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in ASTM D3924. Unless otherwise stated in the test method or paragraph, room temperature shall be 70 ±10 °F (21 ±5 °C) and relative humidity of 50 ±10 percent.

4.3 Qualification inspection. The qualification inspection shall consist of all examinations and tests specified in 4.6.

4.3.1 Qualification inspection samples. The test samples for qualification shall consist of three quarts of the SAE AMS-STD-595, colors 17925 and 36375. The material shall be furnished in the type containers to be used in filling contract orders (see 6.3.1).

4.4 Conformance inspection.

4.4.1 Tests. Conformance inspections shall be performed by the manufacturer on each production batch (see 6.5). Conformance inspection consists of all the tests specified in table II.

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TABLE II. Conformance inspections.

Requirement Paragraph	Test	Test Method Paragraph	ASTM Method
3.5.2	Fineness of grind	---	D1210
3.5.3	Viscosity	---	D1200
3.6.1	Drying time	---	D1640/D1640M/D5895
3.6.2	Surface appearance	4.6.2	---
3.7.1	Color	---	D2244
3.7.2	Gloss	---	D523
3.7.4	Opacity (hiding power)	4.6.5	D2244
3.7.3	Adhesion, tape test	4.6.4	D3359
3.7.5.1 Types III & IV	Impact flexibility	4.6.6.1	D6905
3.7.5.2	Mandrel bend flexibility	4.6.6.2	D522/D522M

4.4.2 Visual inspection of filled containers. Samples shall be selected at random from each lot in accordance with ANSI/ASQ Z1.4, inspection level S-2. The lot size for this examination shall be the number of kits fully prepared for delivery. The samples shall be examined for container fill, weight, and marking.

4.5 Test panels. All test panels shall be prepared under laboratory conditions (see 4.2). With the exception of the panels used for flexibility testing, all test panels shall be aluminum alloy 2024 (T3 temper) conforming to table III, panel A. Panels to be used for flexibility testing shall be aluminum alloy 2024 (O temper) conforming to table III, panel B.

4.5.1 Panel preparation. Test panels shall be prepared as specified in table III.

TABLE III. Aluminum test panels.

Panel	Substrate	Common name	Panel dimensions	Pretreatment
A	SAE AMS4037 (T3 temper)	Al2024-T3 (T3 temper)	0.025 by 3.0 by 6.0 in. (0.625 by 76.2 by 152.4 mm)	MIL-DTL-5541, class 1A (conversion coating)
B	SAE AMS4035 (O temper)	Al2024-O (O temper)	0.020 by 3.0 by 6.0 in. (0.508 by 76.2 by 152.4 mm)	MIL-PRF-8625, type I or IC (anodize, unsealed)

4.5.2 Application of coatings.

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4.5.2.1 Primer coating. Spray apply one cross-coat of a primer coating to a dry film thickness of 0.6 to 0.9 mils (15 to 23 µm) in accordance with ASTM D823, Method D that conforms to MIL-PRF-23377 or MIL-PRF-85582 and air dry for at least five hours or at least two hours respectively.

4.5.2.2 Topcoat. Prepare the coating in accordance with the manufacturer's instructions (the admixed topcoat shall be reduced with thinner conforming to MIL-DTL-81772 if so instructed by the manufacturer). If reducing with thinner, do not exceed the maximum allowable VOC content (see 3.3.1). Spray-apply the coating in accordance with ASTM D823 to a dry film thickness of 1.7 to 2.3 mils (43 to 58 µm). Prior to testing, the applied coating shall be air-dried for a minimum of 14 days at room temperature. For conformance testing, the panels can be air-dried for no less than 24 hours followed by 24 hours at 150 °F.

4.6 Test methods. The tests of this specification shall be conducted in accordance with table IV and 4.6.1 through 4.6.13 with test panels prepared in accordance with 4.5 through 4.5.2. Unless otherwise specified in the test method or paragraph, all testing shall be conducted under the inspection conditions specified in 4.2.

TABLE IV. Test methods.

Test	Requirement Paragraph	Test Paragraph	ASTM Method
Toxicity	3.2	4.6.12	---
Composition	3.3	---	---
Cadmium and lead content	3.2	---	D3335
Hexavalent chromium content	3.2	---	D3718
VOC Content	3.3.1	---	D3960
Thinner Compatibility	3.3.2	---	---
Storage Stability	3.4.1	4.6 & 6.4.3	---
Accelerated Storage Stability	3.4.2	4.6.1	---
Condition in Container	3.5.1	4.6.13	---
Fineness of Grind	3.5.2	---	D1210
Viscosity	3.5.3	---	D1200
Working Properties	3.6	---	---
Drying Time	3.6.1	---	D1640/D1640M/D5895
Surface Appearance	3.6.2	4.6.2	---
Coating system compatibility	3.6.3	---	---
Color	3.7.1	---	D2244
Specular Gloss	3.7.2	4.6.3	D523
Adhesion	3.7.3	4.6.4	D3359
Opacity	3.7.4	4.6.5	D2805
Impact flexibility, Types III & IV	3.7.5.1	4.6.6.1	D6905

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TABLE IV. Test methods – Continued.

Test	Requirement Paragraph	Test Paragraph	ASTM Method
Mandrel bend flexibility	3.7.5.2	4.6.6.2	D522/D522M
Fluid Resistance	3.8.1	4.6.7	---
Weather Resistance	3.8.2	4.6.8	G155
Heat Resistance	3.8.3	4.6.9	D2244
Solvent Resistance (cure) Types III & IV	3.8.4	4.6.10	D5402
Water Resistance	3.8.5	4.6.11	D1308
Identification of material	3.9	---	---

4.6.1 Accelerated storage stability. The coating, in its original, unopened container, shall be stored at 140 ±5 °F (60 ±3 °C) for a minimum of 14 consecutive days. During the storage period, the container shall be placed in a larger, vented container to confine any splashing that may occur if the lid of the unopened container is blown off by increased internal pressure. At the end of the 14-day period, remove the container and allow it to cool to room temperature. If the unopened container is deformed upon removal from storage, do not open and discard. If it is not deformed, open the container carefully and examine its contents for conformance to 3.4.2.

4.6.2 Surface appearance. The coatings shall be applied to test panels in accordance with 4.5 through 4.5.2 for conformance to 3.6.2.

4.6.3 Specular gloss. The coatings shall be applied directly to the unprimed test panels and air-dried for 14 days. Examine in accordance with ASTM D523 at a 60° geometry for conformance to 3.7.2.

4.6.4 Adhesion. The topcoat shall be applied to the test panels in accordance with 4.5 through 4.5.2. The test panels shall be immersed in distilled water for no less than 24 hours at room temperature (see 4.2). Remove the test panels from the water, and within 3 minutes after removal from the water, wipe dry and make two parallel scribes with a stylus through the topcoat to the substrate. The scribes shall be 3/4 of an inch apart and 2-inches long. The panels shall then be scribed to the substrate from opposing ends of the parallel scribes to form an “X” as shown on figure 1. Immediately apply a 1-inch-wide strip of masking tape with the adhesive side down across the scribes. The tape shall have an adhesion of at least 60-ounces/inch widths in accordance with SAE AMS3721, or equivalent, as approved by the qualifying activity, (the tape shelf life is typically one year from date of manufacture). Press the tape against the surface of the topcoat by passing a 4½-pound rubber covered roller, approximately 3½ inches in diameter and 1¾ inches in width across the tape eight times. Remove the tape with one quick motion and examine for topcoat damage. Examine the topcoat for conformance to 3.7.3 and 3.4.2.

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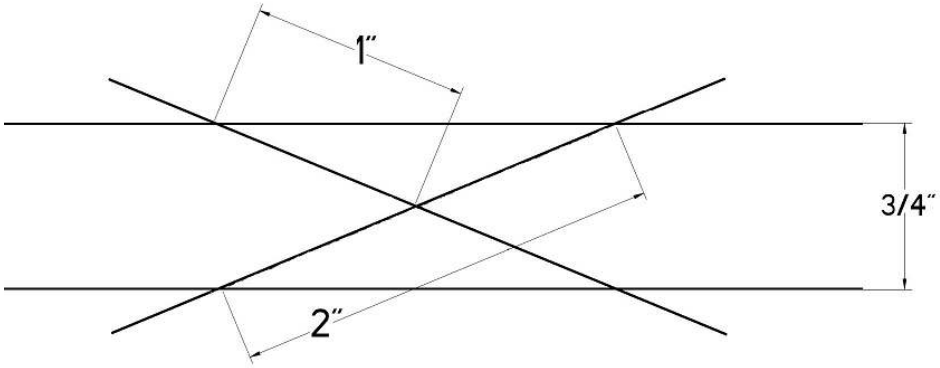


FIGURE 1. Design of X scribe.

4.6.5 Opacity (hiding power). The mixed/admixed topcoat shall be cast to a dry film thickness of 1.9 to 2.1 mil [48 to 53 micrometer (μm)] on a black and white chart (Lenata Form 3B, or equivalent). The contrast ratio shall be determined by measuring the "L" reflectance value of topcoat over the black and white side of the chart, respectively. Determine the "L" value of the topcoat covering the white and black part of the chart, L_W and L_B respectively, in accordance with ASTM D2244. The contrast ratio is calculated as follows: $C = L_B \div L_W$.

4.6.6 Flexibility. Test panels shall be as specified in table III, panel B. Apply the topcoat, without a primer, in accordance with 4.5.2.2, and allow the test panels to air-dry for no less than 14 days before testing. For conformance testing, the panels can be air-dried for no less than 24 hours followed by 24 hours at 150 °F.

4.6.6.1 Impact flexibility (type III and IV only). Test panels shall be prepared in accordance with table III, panel B. The flexibility of the topcoat shall then be tested in accordance with ASTM D6905, using a Gardco GE Universal Impact Tester, Model #172 (or equivalent as approved by the qualifying activity). Place the coated panel, film side downward, on the rubber pad at the bottom of the impacter guide. Drop the impacter on the panel through the impacter guide, ensuring that the impression of the entire rim of the impacter is made in the panel (recommended height of the impacter is 34 inches). Reverse the impacter ends and drop it through the guide on the panel adjacent to the first area of impact. Using 10 power magnification, examine for conformance to 3.7.5.1; record the percent elongation corresponding to the largest spherical impression at which no cracking occurs.

4.6.6.2 Mandrel bend flexibility. Test panels, prepared as specified in table III, panel B, shall be tested in accordance with ASTM D522/D522M, method B. The test shall be performed at a temperature of 24 ±3 °C (75 ±5 °F) using a 1.0 inch mandrel for gloss and semi-gloss colors and a 2.0 inch mandrel for camouflage colors. After testing, examine the topcoat using the unaided eye, for conformance to 3.7.5.2.

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4.6.7 Fluid resistance. When immersed in the fluids in accordance with table V the coatings shall be examined for conformance to 3.8.1.

TABLE V. Fluid immersion conditions.

Fluid	Fluid Temperature	Time of Immersion	
Lubricating oil conforming to MIL-PRF-23699	250 ±5 °F (121 ±3 °C)	Type I Type II Type III & IV	4 hours 4 hours 24 hours
Hydraulic fluid conforming to MIL-PRF-83282	150 ±3 °F (66 ±3 °C)	24 hours	
JP-5 fuel conforming to MIL-DTL-5624	Room temperature	7 days	

4.6.8 Weather resistance. For types I and II the topcoat shall be applied to test panels in accordance with 4.5 through 4.5.2, and tested in accordance with ASTM G155 with a Xenon-arc weatherometer using a daylight filter (Atlas Material Testing Technology or equivalent as approved by the qualifying activity) that is cycling between 102 minutes of light only and 18 minutes of light and water spray for not less than 500 hours.

For types III and IV the topcoat shall be applied to test panels in accordance with 4.5 through 4.5.2, and tested in accordance with ASTM G155 with a Xenon-arc weatherometer using a daylight filter (Atlas Material Testing Technology Company or equivalent as approved by the qualifying activity) that is cycling between 102 minutes of light only and 18 minutes of light and water spray for not less than 1000 hours.

The following conditions shall apply when testing in accordance with ASTM-G155:

a. Black body temperature in cabinet: 145 ±5 °F (63 ±3 °C).
b. Relative humidity in cabinet: 50 ±5 percent.
c. Intensity (spectral irradiance) of Xenon-arc: 0.35 ±0.01 Watts/meter^2 (W/m^2) at a wavelength of 340 nm.
d. Cabinet temperature: 108 ±5 °F (42 ±3 °C).

4.6.9 Heat resistance. Prepare three test panels in accordance with 4.5 through 4.5.2. One test panel shall act as control, the other two as the test specimens. Expose the test specimens to 121 ±3 °C (250 ±5 °F) for 60 minutes. After exposure, allow the test panels to cool to room temperature. The topcoat shall then be tested in accordance with ASTM D2244 using CIELAB color coordinates with a 10° observer and D65 illuminant and examined for conformance to 3.8.3.

4.6.10 Solvent resistance (cure), (type III and IV). Test panels shall be prepared in accordance with 4.5 through 4.5.2. A cotton, terrycloth rag shall be soaked in MEK solvent and rubbed back and forth 25 times (50 passes) over the topcoat with firm finger pressure in accordance with ASTM D5402 method A. The topcoat shall be examined for conformance to 3.8.4.

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4.6.11 Water resistance. Test panels shall be prepared in accordance with 4.5 through 4.5.2, with the exception that the applied coatings shall be air-dried for seven days at room temperature. The test panels shall then be immersed in distilled water in accordance with ASTM D1308 for 24 hours at room temperature. Within five minutes of removal from the water, the coatings shall be examined for conformance to 3.8.5.

4.6.12 Toxicity. The product shall be evaluated by the Navy Environmental Health Center (NAVENVIRHLTHCEN) using the administrative Health Hazard Assessment (HHA) (see 6.4.1).

4.6.13 Condition in container. Upon opening a full previously unopened container the coating shall be examined for conformance to 3.5.1. The coating shall then be mixed with a hand paddle and examined for conformance to 3.5.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful but is not mandatory.)

6.1 Intended use. MIL-PRF-81352 Type I and II are VOC compliant (340 g/l maximum) single component coatings that can be used for application of temporary markings/MODEX. These lacquer coatings dry by solvent evaporation. For the application of temporary markings, these lacquer coatings are applied over clean, lightly scuff sanded and solvent wiped MIL-PRF-85285 topcoats. Lacquer coatings do not provide sufficient durability for use as a touch-up material for conventional coatings.

MIL-PRF-81352 Type III and IV coatings are more robust than type I and II coatings and can be used for general topcoat repair. The type III and IV coatings are single component, user friendly (requiring no proper mixing ratio) and are VOC compliant. Type III and IV coatings are intended to be an exterior coating for touch-up purposes over epoxy primer conforming to MIL-PRF-23377 or MIL-PRF-85582. Type III topcoats are polyurethanes that contain isocyanates and require humidity for curing. Type IV topcoats are isocyanate free moisture cured polysiloxane topcoats.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification, including any amendments.
b. Type and color (see 1.2 and 1.3).
c. Quantity, size, and type of containers in which the coating is to be furnished (see 1.3).
d. Packaging requirements (see 5.1).
e. Test or inspection report, if required (see 6.7).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion on the Qualified Products List QPL-81352 in the Qualified Products Database (QPD). The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Commander, Naval Air Warfare Center Aircraft Division, Code AB32000, Building 2188, Patuxent River, MD 20670-1908. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification inspection sample identification. Qualification inspection samples are to be forwarded to the laboratory designated in the letter of authorization (see 6.3) and identified as follows:

Qualification test samples.
Specification MIL-PRF-81352, Type____, Color____
"Coatings, Aircraft Touch-up"
Manufacturer's name and product number
Submitted by (name and date) for qualification testing in accordance with authorization (reference authorizing letter)

6.3.2 Retention of qualification. To retain qualification of products approved for listing on the Qualified Products Database (QPD), one of the items that the manufacturer will be requested to verify by certification via DD Form 1718 *Certification of Qualified Products* to the qualifying activity is that its product(s) complies with the requirements of this specification. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification will be in two-year intervals from the date of original qualification and will be initiated by the qualifying activity. Qualification retention information can be found in the SD-6 *Provisions of Qualification.*

6.3.3 Manufacturer's advertising. The following statement serves as a precaution of any misrepresentation of products procured to this specification. It also informs manufacturers of the qualification requirement for the advertisement of qualified products to this specification.

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Manufacturers are prohibited from displaying this specification on any product packaging, labeling, technical data sheets or anywhere else in reference to a product that has not been authorized by the qualifying activity on the applicable QPL in the QPD. See DoDM 4120.24 *Defense Standardization Program (DSP) Procedures – Enclosure 14* for additional information on manufacturer's advertising.

6.4 Material hazards.

6.4.1 Toxicity evaluation. Health Hazard Assessment (HHA) is a review of a product based on information submitted by the manufacturer at the request of the qualifying activity to assess health hazards associated with the handling, application, use and removal of a product. Sufficient data to permit an HHA of a product should be provided by the manufacturer/distributor and submitted to the Navy and Marine Corps Public Health Center (NAVMCPUBHLTHCEN) Industrial Hygiene Department, usn.hampton-roads.navmcpubhlthcenpors.list.nmcphc-HHA@mail.mil or Commanding Officer, Navy and Marine Corps Public Health Center, ATTN: Industrial Hygiene-HHA, 620 John Paul Jones Circle, Portsmouth, VA 20378-2103.

To obtain current technical information requirements specified by the NAVMCPUBHLTHCEN or forward any questions concerning toxicity or information required to conduct an HHA, contact: Head, Acquisition Technical Support Division, (757) 953-0978; DSN 377-0978 usn.hampton-roads.navmcpubhlthcenpors.list.nmcphc-HHA@mail.mil.

6.4.2 Safety Data Sheet (SDS). The 29 CFR 1910.1200 requires that an SDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers should identify the activities requiring copies of an SDS.

6.4.3 Storage stability, inspection, and other information. In addition to the qualification test samples, the qualifying activity will request the manufacturer to submit to the qualification activity: (a) certification that the manufacturer's material meets the extended storage stability requirements (see 3.4.1); (b) one copy of the SDS (see 6.4.2); (c) a certified test report showing that the material conforms to the requirements of this specification; and (d) certification that the following chemicals were not used in the formulation of this topcoat: methylene chloride, trichloroethane and trichlorotrifluoroethane; € in the case of type IV, certification that product contains no isocyanates.

6.5 Batch and lot formation. A batch consists of all topcoat material manufactured during one continuous operation and forming part of one contract or order for delivery. A lot consists of all topcoats of the same color, manufactured at one time from one batch, forming part of one contract, and submitted for acceptance.

6.6 Safely handling MEK solvent. To minimize exposure to MEK solvent, it is recommended that personnel conducting the solvent resistance (cure) test, as a minimum, wear either butyl rubber or Teflon gloves and a National Institute of Occupational Safety and Health (NIOSH) approved half-face respirator equipped with organic vapor cartridges and goggles, or a full-face respirator equipped with organic vapor cartridges.

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6.7 Report of tests. When required (see 6.2), the manufacturer must submit to the procuring activity a certified test report for each lot of material. The test report must show the results of the conformance tests of this specification.

6.8 Subject term (keyword) listing.

Acrylic
Alkyd
Polyurethane
Polysiloxane

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MR
Navy – AS
Air Force – 20
GSA – FAS

Preparing activity:
Navy – AS
(Project 8010-2024-003)

Review activities:
Army – MD1
Navy – CG

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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